

Project Planning Report: Optimisation Methods for Variational Inference

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Background and Problem Intuition

Bayesian inference provides principled uncertainty quantification but requires computing intractable posterior distributions. For most real-world models, the posterior integral,

$$p(\boldsymbol{\theta} \mid \mathbf{y}) = \frac{p(\mathbf{y} \mid \boldsymbol{\theta})p(\boldsymbol{\theta})}{p(\mathbf{y})},$$

cannot be evaluated analytically. Variational Inference (VI) converts this inference problem into an optimisation problem, making it directly amenable to the gradient-based methods in this course.

Instead of computing the true posterior, VI finds the “best” approximation $q(\boldsymbol{\theta})$ from a tractable family by maximising the Evidence Lower Bound (ELBO). This transforms intractable integration into tractable optimisation—I think this is a natural fit for an optimisation course project.

Key Notation

Symbol	Meaning
$\boldsymbol{\beta}$	Regression coefficient vector
$\mathbf{u}$	Random effects vector (hierarchical model)
$\boldsymbol{\varepsilon}$	Observation noise vector
τ_e	Observation-noise precision
τ_u	Random-effects precision
$\boldsymbol{\mu}_\beta$	Mean of variational posterior $q(\boldsymbol{\beta})$
$\boldsymbol{\Sigma}_\beta$	Covariance of variational posterior $q(\boldsymbol{\beta})$
$\boldsymbol{\phi}$	Variational parameter vector $(\boldsymbol{\mu}_\beta, \boldsymbol{\Sigma}_\beta, a_e, b_e)$
$h(\alpha)$	Line search slice ELBO($\boldsymbol{\phi}^{(t)} + \alpha \mathbf{d}^{(t)}$); replaces course notation $\phi(\alpha)$

Optimisation Problem Formulation

Decision Variables

For Bayesian linear regression with p predictors, the variational parameters (decision variables) are:

- $\boldsymbol{\mu}_\beta \in \mathbb{R}^p$: Posterior mean of regression coefficients
- $\boldsymbol{\Sigma}_\beta \in \mathbb{R}^{p \times p}$: Posterior covariance (symmetric positive definite)
- $a_e, b_e > 0$: Shape and rate parameters of Gamma distribution for precision

Total dimension: $d = p + \frac{p(p+1)}{2} + 2 = 11$ variables for $p = 3$ predictors.

Objective Function

Maximise the Evidence Lower Bound (ELBO):

$$\text{ELBO}(\boldsymbol{\phi}) = \mathbb{E}_q[\log p(\mathbf{y}, \boldsymbol{\beta}, \tau_e)] - \mathbb{E}_q[\log q(\boldsymbol{\beta}, \tau_e)] \quad (1)$$

where $\boldsymbol{\phi} = (\boldsymbol{\mu}_\beta, \boldsymbol{\Sigma}_\beta, a_e, b_e)$ are the variational parameters.

Constraints

- $\boldsymbol{\Sigma}_\beta \succ 0$: Covariance matrix must be positive definite
- $a_e, b_e > 0$: Gamma parameters must be positive
- Optional: Minimum variance constraints $\sigma_{\beta_j}^2 \geq \sigma_{\min}^2$ to prevent underdispersion

These constraints are handled via reparameterisation, as described in the following subsections.

Constraint Handling via Reparameterisation

The variational parameters are not free. $\boldsymbol{\Sigma}_\beta$ must remain symmetric positive definite throughout optimisation, and a_e, b_e must remain strictly positive. Four strategies exist for handling these constraints in a gradient-based optimiser: ignore them (the algorithm crashes when $\log |\boldsymbol{\Sigma}_\beta|$ becomes undefined); add a penalty term (introduces a tuning parameter and ill-conditioning as $\rho \rightarrow \infty$); project back to the feasible set after each step (costs $O(p^3)$ per iteration and is non-differentiable at the boundary, breaking BFGS); or reparameterise so the constraints are encoded structurally and cannot be violated. This work uses reparameterisation throughout—it requires no tuning parameters, no outer loops, and produces a smooth unconstrained problem fully compatible with gradient ascent, Newton, and BFGS. Two transformations are applied: one for the matrix constraint on $\boldsymbol{\Sigma}_\beta$, and one for the scalar positivity constraints on a_e and b_e .

Cholesky Reparameterisation

$\boldsymbol{\Sigma}_\beta$ is the covariance matrix of the variational distribution $q(\boldsymbol{\beta}) = \mathcal{N}(\boldsymbol{\mu}_\beta, \boldsymbol{\Sigma}_\beta)$, and must therefore remain symmetric positive definite (SPD) at every iteration. A gradient step has no awareness of this constraint: nothing stops an update from producing a matrix with a negative eigenvalue, after which $\log |\boldsymbol{\Sigma}_\beta|$ and $\boldsymbol{\Sigma}_\beta^{-1}$ are undefined and the algorithm crashes.

The transformation is $\Sigma_\beta = \mathbf{L}\mathbf{L}^T$, where $\mathbf{L}$ is lower triangular. Optimising over $\mathbf{L}$ instead of Σ_β directly ensures any $\mathbf{L}$ produces a valid symmetric positive semidefinite matrix by construction. However, a residual constraint remains: the diagonal entries ℓ_{ii} must be strictly positive for Σ_β to be positive definite rather than merely positive semidefinite. Three options exist: ignore the diagonal constraint (positivity can still be violated, recovering the original problem); apply a softplus function $\ell_{ii} = \log(1 + e^{u_i})$ (valid but less standard); or apply a log-space transformation $\ell_{ii} = e^{\tilde{\ell}_{ii}}$, keeping $\tilde{\ell}_{ii} \in \mathbb{R}$ unconstrained. The log-space option is used here. The parameter count is unchanged at $p(p+1)/2$ free entries: p log-diagonal entries $\tilde{\ell}_{ii}$ and $p(p-1)/2$ unconstrained off-diagonal entries ℓ_{ij} , $i > j$.

Log-Space Reparameterisation

The Gamma shape and rate parameters $a_e, b_e > 0$ carry a scalar positivity constraint. The same reparameterisation principle applies: define $\eta_a = \log a_e$ and $\eta_b = \log b_e$, so that $a_e = e^{\eta_a}$ and $b_e = e^{\eta_b}$. Since the exponential function maps $\mathbb{R}$ onto $(0, \infty)$, the optimiser moves η_a, η_b freely over the real line and the positivity constraint is structurally impossible to violate. This is the scalar analogue of the Cholesky diagonal transformation, and the same principle as substituting $x = e^t$ to eliminate $x > 0$ in single-variable optimisation.

After both transformations, the effective optimisation variables are $(\boldsymbol{\mu}_\beta, \tilde{\ell}_{11}, \ell_{21}, \tilde{\ell}_{22}, \dots, \eta_a, \eta_b)$ —all unconstrained reals. The quantities Σ_β , a_e , b_e are recovered on demand when evaluating the ELBO and its gradients.

Computational Characteristics

- Gradient computation: $O(np^2)$ per evaluation (dominated by matrix operations)
- Hessian computation: $O(np^2 + d^3)$ for Newton’s method
- Parameters exhibit coupling (changing $\boldsymbol{\mu}_\beta$ affects optimal Σ_β)
- Must handle special functions: digamma $\psi(\cdot)$, trigamma $\psi'(\cdot)$ for Gamma distribution

Test Case: Bayesian Linear Regression

The test case uses Bayesian linear regression with:

$$\mathbf{y} \sim \mathcal{N}(\mathbf{X}\boldsymbol{\beta}, \tau_e^{-1}\mathbf{I}_n) \quad (\text{likelihood}) \quad (2)$$

$$\boldsymbol{\beta} \sim \mathcal{N}(\mathbf{0}, \lambda_\beta^{-1}\mathbf{I}_p) \quad (\text{prior on coefficients}) \quad (3)$$

$$\tau_e \sim \text{Gamma}(\alpha_e, \gamma_e) \quad (\text{prior on precision}) \quad (4)$$

Mean-field approximation:

$$q(\boldsymbol{\beta}, \tau_e) = q(\boldsymbol{\beta}) q(\tau_e) \quad (5)$$

$$q(\boldsymbol{\beta}) = \mathcal{N}(\boldsymbol{\mu}_\beta, \boldsymbol{\Sigma}_\beta) \quad (6)$$

$$q(\tau_e) = \text{Gamma}(a_e, b_e) \quad (7)$$

Variational parameters: $\phi = (\boldsymbol{\mu}_\beta, \boldsymbol{\Sigma}_\beta, a_e, b_e)$ with dimension $d = p + \frac{p(p+1)}{2} + 2$
For $p = 3$ predictors: $d = 3 + 6 + 2 = 11$ parameters to optimise.

Overview of Optimisation Methods

Methods considered

- **CAVI:** Domain-specific baseline exploiting problem structure
- **Gradient Ascent:** Baseline first-order method, expected to be slowest but most robust
- **Newton’s Method:** Best convergence rate (quadratic) if Hessian well-conditioned
- **BFGS:** “Best general-purpose optimiser” for smooth problems (course recommendation)

Baseline Method: CAVI¹

Coordinate Ascent Variational Inference (CAVI) is VI-specific. It updates each variational parameter block sequentially using closed-form solutions:

$$\boldsymbol{\Sigma}_\beta \leftarrow \left(\frac{a_e}{b_e} \mathbf{X}^T \mathbf{X} \right)^{-1} \quad (8)$$

$$\boldsymbol{\mu}_\beta \leftarrow \frac{a_e}{b_e} \boldsymbol{\Sigma}_\beta \mathbf{X}^T \mathbf{y} \quad (9)$$

$$a_e \leftarrow \alpha_e + \frac{n}{2} \quad (10)$$

$$b_e \leftarrow \gamma_e + \frac{1}{2} [\|\mathbf{y} - \mathbf{X}\boldsymbol{\mu}_\beta\|^2 + \text{tr}(\mathbf{X}^T \mathbf{X} \boldsymbol{\Sigma}_\beta)] \quad (11)$$

Properties: Monotonic ELBO increase ², linear convergence, $O(np^2)$ per iteration. Provides analytical benchmark for validation.

¹Blei, Kucukelbir, and McAuliffe (2017), Variational Inference: A Review for Statisticians (Journal of the American Statistical Association).

²Monotonic ELBO increase means the objective is non-decreasing across iterations: $\mathcal{L}^{(t+1)} \geq \mathcal{L}^{(t)}$. In CAVI, each coordinate update maximises the ELBO with respect to one variational factor while holding the others fixed, so each step cannot decrease the ELBO.

Mapping ENEL445 Methods to VI Problem

1. Parameter Ascent (Course: gradient descent)

- Standard iterative update: $\phi^{(t+1)} = \phi^{(t)} + \alpha_t \nabla \text{ELBO}(\phi^{(t)})$
- Line search from Chapter 3.2 (bisection/Wolfe conditions)³
- Stopping criteria: $\|\nabla \text{ELBO}\| < \epsilon$
- Convergence: Linear rate

2. Newton's Method

- Standard Newton update: $\phi^{(t+1)} = \phi^{(t)} + \alpha_t [\nabla^2 \text{ELBO}]^{-1} \nabla \text{ELBO}$
- Requires computing and inverting the Hessian matrix
- Computational cost: $O(d^3)$ per iteration for Hessian inversion
- Convergence: Quadratic rate near optimum
- Challenge: Hessian may not be positive definite away from the optimum

3. BFGS Quasi-Newton (one of five required methods from Project 2)

- BFGS update formula builds inverse Hessian approximation
- Uses gradient differences: $\mathbf{s}_k = \phi^{(k)} - \phi^{(k-1)}$, $\mathbf{y}_k = \nabla \text{ELBO}(\phi^{(k)}) - \nabla \text{ELBO}(\phi^{(k-1)})$
- Woodbury identity for efficient updates of inverse Hessian
- Computational cost: $O(d^2)$ per iteration
- Convergence: Superlinear rate
- Widely regarded as best general-purpose optimiser for smooth unconstrained problems

Comparison will reveal whether general optimisation methods can compete with or outperform problem-specific algorithms

Solution Plan

Phase 1: Data Generation and Baseline (Completed)

Data simulation:

- Generate $n = 100$ observations, $p = 3$ predictors
- Design matrix $\mathbf{X} \sim \mathcal{N}(0, 1)$, true coefficients $\beta_{\text{true}} = [1, -0.5, 2]^T$
- Noise: $\epsilon \sim \mathcal{N}(0, 0.5^2)$

Mathematical foundations (in progress):

- Complete ELBO derivation for Bayesian linear regression
- CAVI closed-form update equations

³The ENEL445 course uses $\phi(\alpha)$ for the line search slice. Since ϕ is reserved here for the variational parameter vector, we write $h(\alpha) = \text{ELBO}(\phi^{(t)} + \alpha \mathbf{d}^{(t)})$ instead, with $h'(\alpha) = \nabla \text{ELBO}(\phi^{(t)} + \alpha \mathbf{d}^{(t)})^T \mathbf{d}^{(t)}$.

- Gradient expressions: ∇_{μ_β} ELBO, ∇_{Σ_β} ELBO, ∇_{a_e} ELBO, ∇_{b_e} ELBO
- Hessian block structure: diagonal and mixed blocks
- BFGS update formulation with Woodbury identity

Derivations to be documented in `vi_optimisation_solution.pdf` (or an appendix to the report?)

Phase 2: Implementation (In Progress)

1. Data Generation (5%)

- Simulate data: $n = 100$ observations, $p = 3$ predictors
- Generate $\mathbf{X}$ from standard normal, $\beta_{\text{true}} = [1, -0.5, 2]^T$
- Add noise: $\epsilon \sim \mathcal{N}(0, 0.5^2)$

2. Baseline Implementation: CAVI (15%)

- Implement closed-form coordinate updates
- Track ELBO convergence
- Stopping criterion: $|\text{ELBO}^{(t+1)} - \text{ELBO}^{(t)}| < 10^{-6}$

3. Method 1: Gradient Ascent (25%)

- Implement ELBO gradient computation
- Line search: bisection method or Wolfe conditions
- Stopping criterion: $\|\nabla \text{ELBO}\| < 10^{-6}$
- Track iteration count and ELBO values

4. Method 2: Newton's Method (0%)

- Implement Hessian computation (block structure)
- Regularization: add $\lambda \mathbf{I}$ if Hessian not positive definite
- Line search for step size α_t
- Track convergence and computational cost

5. Method 3: BFGS Quasi-Newton (0%)

- Initialise inverse Hessian approximation: $\mathbf{B}_0 = \mathbf{I}_d$
- Implement BFGS update formula with Woodbury identity
- Line search with Wolfe conditions
- Track superlinear convergence

6. Comparison and Analysis (5%)

- Compare convergence speed (iterations to 10^{-6} tolerance)
- Compare computational cost (time per iteration, total time)
- Compare solution quality (posterior approximation accuracy)
- Analyse when each method performs best

Success Metrics

1. **Convergence:** All methods reach ELBO optimum within tolerance
2. **Correctness:**
 - CAVI correctness: Gradient/Newton/BFGS implementation is correct if it reaches the same ELBO optimum / variational fixed point as CAVI (same VI objective, same variational family)
 - MCMC correctness: VI is assessed against MCMC as a posterior-accuracy benchmark (means, intervals, calibration), not expected to match exactly because VI is approximate
3. **Performance hierarchy:**
 - Expected optimisation speed: BFGS > Newton > Gradient Ascent (for ELBO convergence)
 - CAVI likely fastest for this conjugate problem
 - MCMC/Gibbs expected slowest computationally, but best for posterior uncertainty accuracy (gold-standard benchmark, not a direct ELBO-optimiser competitor)
4. **Understanding:** Clear explanation of why each method performs as it does

Progress To Date

Work in progress using an iterative / Agile method

1. **Problem formulation:** Draft optimisation problem specified with decision variables, objective function (ELBO), and constraints
2. **Mathematical derivations**⁴:
 - Draft ELBO derivation from first principles
 - Draft CAVI algorithm with closed-form updates
 - Analytical gradient expressions for all 11 variational parameters (unverified)
 - Potential Hessian block computations for Newton’s method
 - BFGS quasi-Newton formulation
 - Attempt at Penalty method formulation for constrained optimisation
3. **Draft method selection justification:**
 - Parameter ascent $\leftrightarrow$ Algorithm 3.2 (gradient descent)
 - Newton’s method $\leftrightarrow$ Equation 3.26
 - BFGS $\leftrightarrow$ Equations 4.50-4.51 (one of five required project methods⁵)
4. **Validation strategy:** Established CAVI as analytical benchmark for correctness checking

Preliminary Results

Implementation is pending, the mathematical analysis hints⁶

- **Gradient structure:** $\nabla_{\mu_\beta} \text{ELBO} \propto \mathbf{X}^T(\mathbf{y} - \mathbf{X}\mu_\beta)$ resembles least-squares gradient, suggesting ELBO surface may be well-behaved
- **Hessian properties:** Diagonal blocks are negative definite (ELBO is locally concave?), supporting Newton’s method applicability
- **Dimensionality:** With $d = 11$ parameters, Hessian inversion costs may be manageable for Newton’s method (L-BFGS reduction may not be needed)
- **Parameter coupling:** Mixed Hessian blocks non-zero, indicating gradient-based methods exploiting joint updates should outperform coordinate ascent

⁴This needs some kind of peer review

⁵Project 2 specification

⁶I haven’t been over this enough to make any claim of completeness

Next Steps

1. Implement CAVI baseline in Python (Week 7)
2. Implement parameter ascent with line search (Week 7)
3. Implement Newton's method with regularization (Week 8)
4. Implement BFGS quasi-Newton (Week 8)
5. Monte Carlo comparison: convergence iterations, wall-clock time, ELBO values (Week 8-9)
6. Final report with performance analysis (Week 12)

Alignment with Course Requirements

This project is looking if the core learning objectives of ENEL445 are potentials:

1. **Optimisation formulation:** Convert inference problem to unconstrained optimisation
2. **Gradient computation:** Analytical derivation and numerical implementation
3. **Method implementation:** Code gradient ascent, Newton, and BFGS from scratch
4. **Line search:** Apply bisection or Wolfe conditions for step size selection
5. **Stopping criteria:** Implement gradient norm and function value convergence tests
6. **Performance comparison:** Empirical evaluation of convergence rates and computational costs
7. **Method selection:** Understand when/which optimisation algorithm is to be used

Key insight: The VI problem represents a real computational challenge in machine learning, making the comparison of optimisation methods both valuable and relevant.

Timeline

March 17, 2026	Mathematical derivations completed
March 18, 2026	Data generation and CAVI implementation
March 19, 2026	Gradient ascent and Newton's method implementation
March 20, 2026	BFGS implementation and comparison analysis
March 20, 2026	Progress report submission

Summary

The Variational Inference optimisation problem provides a testbed for implementing and comparing gradient-based methods, and,

- **Applies to** course material (gradient descent, Newton, BFGS)
- **Requires** implementation of algorithms with line search and stopping criteria
- **Enables** performance comparison on a realistic and meaningful problem
- **Demonstrates** an application of optimisation theory to modern machine learning